



# Enterprise AI Knowledge & Retrieval Platform

A private generative-AI platform for a large diversified holding group — a conversational assistant that answers questions over the organization's entire document corpus with citations, and a board-meeting copilot that prepares, fact-checks, and stress-tests decisions live. Powered by agentic Retrieval-Augmented Generation, a knowledge graph, and a fleet of specialized LLM agents.

DOMAIN

CORE

MODEL

PLATFORM

**Enterprise Generative AI**

**Agentic RAG + Knowledge Graph**

**Private, permissioned SaaS**

**Azure · FastAPI · React**

**Confidentiality.** Sanitized portfolio case study. The client identity, product names, brand assets, proprietary prompts, business rules, model deployment names, and internal source are withheld under NDA. Visuals are original architecture diagrams; no client screenshots, data, or credentials are reproduced. Technique parameters (fusion constants, embedding dimensions, token budgets) are generic; commercial figures are intentionally omitted.

© 2026 Rohail K. Malhi — All rights reserved. Proprietary work; **not open-source**. The frameworks, architecture, and artifacts shown here may not be copied, adapted, or repurposed. Shared solely to evaluate the author's experience.

## AT A GLANCE

### CHALLENGE

A holding group's institutional knowledge was trapped in hundreds of documents. Keyword search returned files, not answers; generic chatbots hallucinate; board meetings ran on human memory; and every answer had to be grounded, cited, and permissioned.

### SOLUTION

A private agentic-RAG platform: hybrid retrieval + rank fusion + two-stage reranking over an authorized corpus, grounded generation with citations, a two-phase-LLM knowledge graph, and a board-meeting copilot with an adversarial decision panel.

### RESULT

Plain-English questions answered with sourced, permission-safe answers; meetings prepared, fact-checked, and stress-tested by AI; and a fully observable, measurable AI system delivered on Azure.

### Hybrid + RRF

Keyword + dense vector + HyDE, fused ( $k = 60$ )

### Grounded + cited

Answer-only-from-authorized-context

### Knowledge graph

Two-phase LLM entity + relation extraction

### Meeting copilot

Advocate / critic / arbiter / risk panel

### CHALLENGE

## Knowledge trapped, trust non-negotiable

A large, diversified holding group sits on an enormous, ever-growing body of institutional knowledge — board packs, financials, policies, legal agreements, rate cards, minutes, and market news — across hundreds of documents and dozens of subsidiaries. Several problems compounded:

- **Answers trapped in documents.** No way to ask a plain-English question and get a trustworthy, *sourced* answer. Keyword search returned files; generic chatbots hallucinate — unacceptable when a wrong number reaches a board.
- **Meetings ran on memory.** Preparing meant reading everything by hand; no way to fact-check a claim, surface a related past decision, or pressure-test a recommendation in the room; capturing outcomes was manual.
- **Knowledge was unconnected.** The relationships that matter — who supplies whom, which risk touches which portfolio, who owns which initiative — lived only in people's heads.
- **Trust, security & governance.** Every answer had to be grounded in *authorized* documents only, per user and per document, with nothing leaking across permission boundaries — and every response traceable and measurable.

**SOLUTION**

## Two products over one agentic-RAG core

A conversational **knowledge assistant** (ask anything, get a grounded, cited answer from authorized documents) and a **board-meeting copilot** (prepare, fact-check, and stress-test decisions), both on one retrieval engine engineered for accuracy and trust — not a thin wrapper around a chat model.

- **Query understanding first.** One unified LLM agent classifies domain, extracts entities, expands keywords, and writes a HyDE hypothetical answer in a single call — consolidating five prior LLM calls into one.
- **Hybrid retrieval + fusion.** Boosted keyword, dense-vector kNN, and HyDE searches run in parallel and merge via Reciprocal Rank Fusion (k=60) — exact-match precision, semantic recall, and vocabulary-gap bridging together.
- **Signal-aware ranking + rerank.** Function-score (recency decay, doc-type priority, entity/domain boosts), then a cross-encoder and an LLM-as-judge contextual reranker refine the final context.
- **Grounded generation.** A document-level authorization prefilter is pushed into the search; an answer-only-from-context mode plus mandatory numbered citations mean the platform answers only from what a user may see, and shows its work.
- **Knowledge graph.** Two-phase LLM extraction (entities, then relationships constrained to them) builds a de-duplicated, synonym-merged graph that both visualizes connected knowledge and sharpens retrieval.
- **Meeting copilot.** Pre-meeting insight chains, live fact-checking and question generation, an advocate/critic/arbitrator/risk decision panel, and post-meeting action items, summaries, and an LLM-scripted TTS podcast.

### Retrieval pipeline at a glance

```
Question + history → Unified query-understanding agent (domain · entities · keywords · HyDE)
Next-action router → retrieve / clarify / summarize / list      DB router → session vs historical
Hybrid retrieval (parallel): keyword (boosted BM25) + dense-vector kNN (3072-dim) + HyDE vector
    → Reciprocal Rank Fusion (k=60) → function-score (recency decay · doc-type · entity/domain boosts)
    → Rerank: cross-encoder → LLM-as-judge (relevance / fit / quality)
Prompt assembly: doc-level auth prefilter · citations · token-budget packing → Grounded GPT-4o answer
```

### Technology

Agentic RAG   Azure OpenAI   GPT-4o   GPT-4o-mini   OpenAI Agents SDK   LangChain   LiteLLM

Hybrid search   HyDE   Reciprocal Rank Fusion   Cross-encoder rerank   LLM-as-judge   3072-dim embeddings

MPNet   Elasticsearch   Knowledge graph   Two-phase extraction   tiktoken   scikit-learn / TF-IDF   NLTK

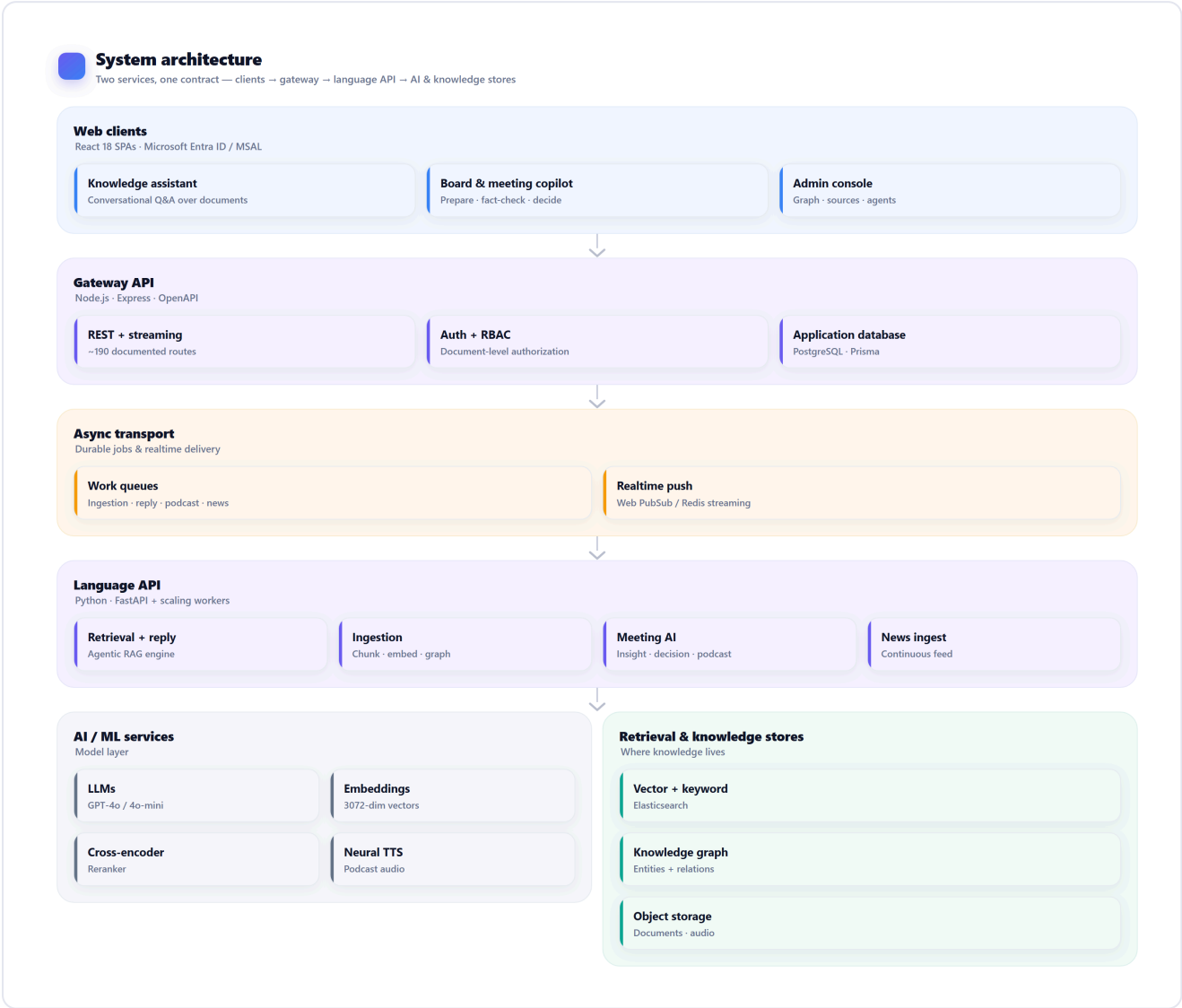
Neural TTS   Prompt-injection guardrail   FastAPI   Python   Node.js / Express   React 18

PostgreSQL / Prisma   Redis   Entra ID / MSAL   OpenTelemetry   Jaeger   Azure Kubernetes Service

THE PLATFORM

System architecture

Two services, one contract — clients to gateway to language service to AI & knowledge stores. Original diagram; no client screenshots or data reproduced.

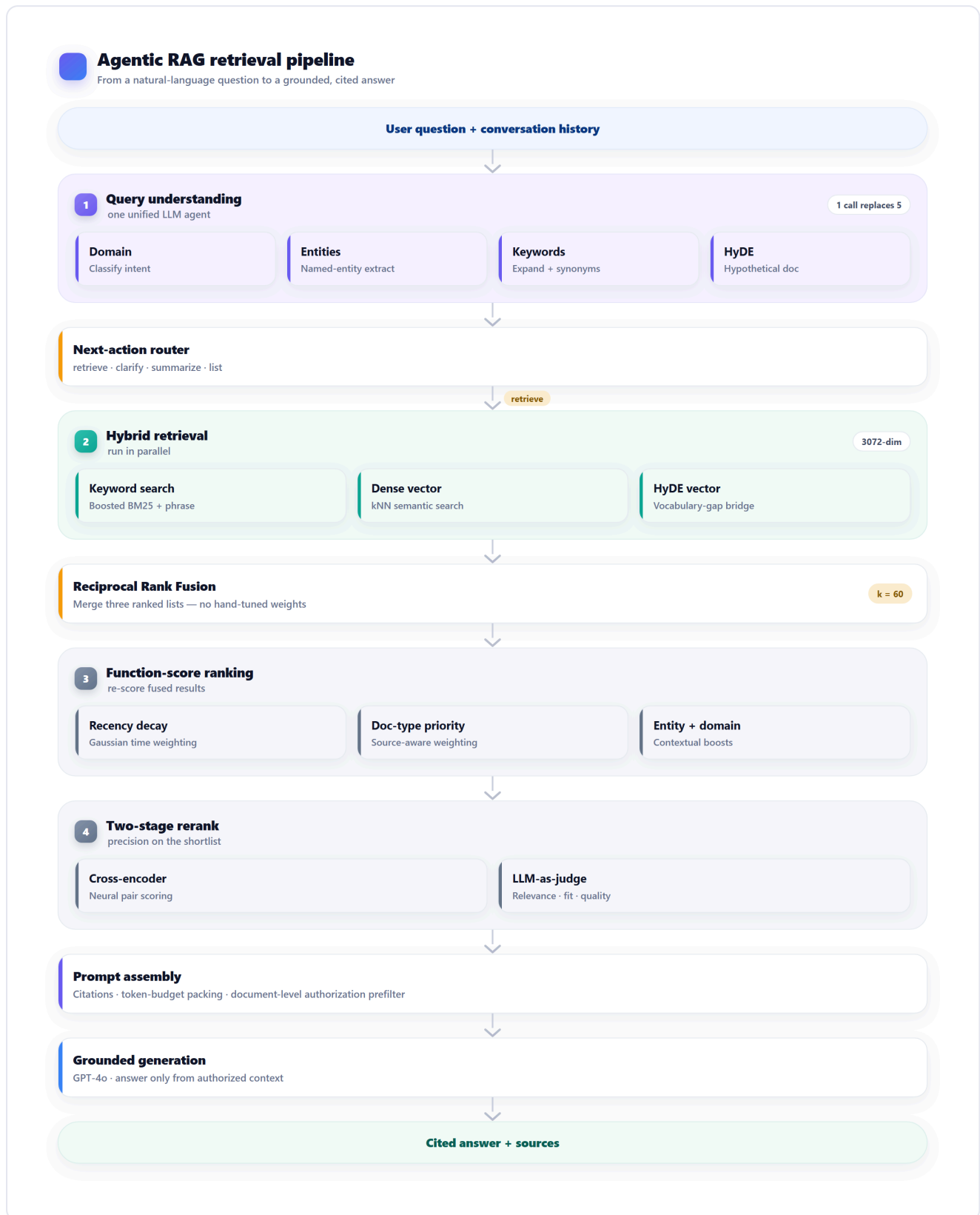


React clients over a Node.js gateway that brokers to a Python/FastAPI language service and its workers via async queues; the AI/ML model layer and the retrieval/knowledge stores sit behind the language service.

## THE CORE

# The agentic-RAG retrieval pipeline

From a natural-language question to a grounded, cited answer.

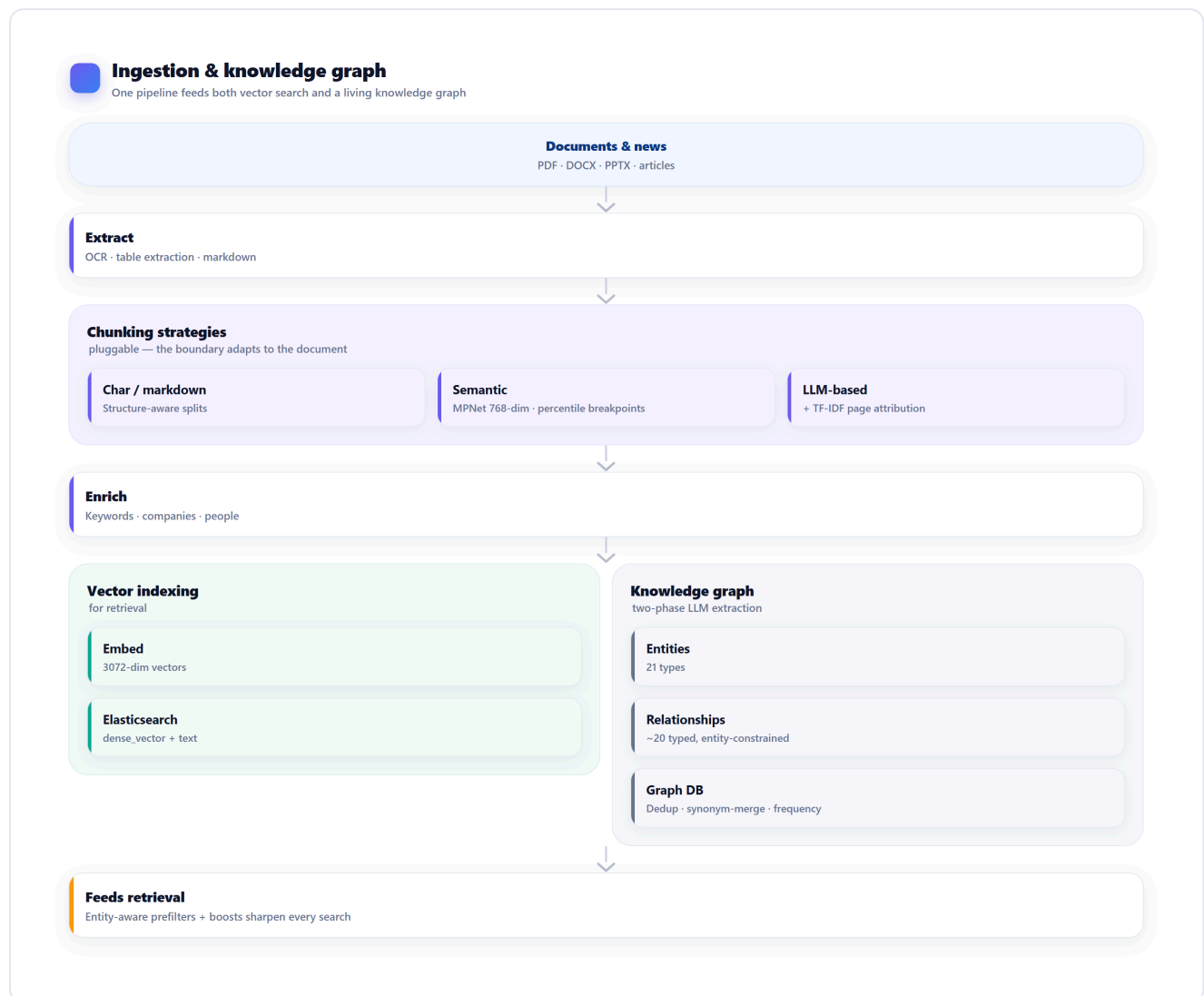


Unified query understanding, action routing, parallel hybrid retrieval, Reciprocal Rank Fusion, function-score ranking, and two-stage reranking before grounded, cited generation — accuracy engineered into every stage.

## INGESTION

# Ingestion & the knowledge graph

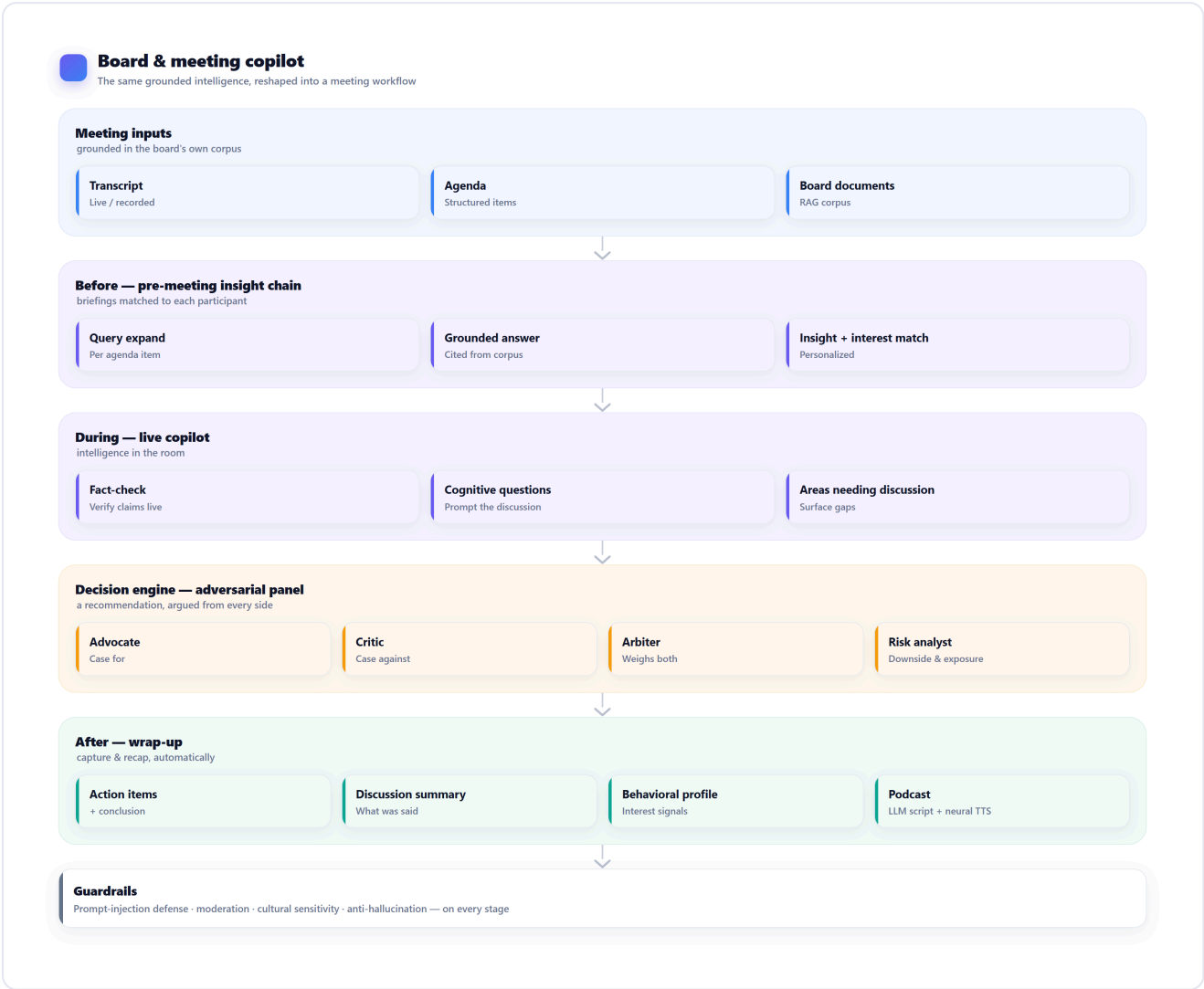
One pipeline feeds both vector search and a living knowledge graph.



Pluggable chunking feeds both a vector index and a two-phase LLM graph extraction; the graph loops back to make retrieval smarter via entity-aware prefilters and boosts.

# The board & meeting copilot

The same grounded intelligence, reshaped into a meeting workflow.



Pre-meeting insight chains, live fact-check and questions, an adversarial advocate/critic/arbiter/risk decision panel, and automatic post-meeting wrap-up — guardrailed on every stage.

**RESULT**

## Trustworthy AI over the whole corpus

### Sourced answers

Grounded, cited, permission-safe responses

### AI in the room

Meetings prepared, fact-checked & stress-tested

### Observable

Tracing + per-response eval metrics

- **Agentic RAG, not a chatbot wrapper.** Query understanding, routing, hybrid retrieval, fusion, ranking, two-stage reranking, budgeted assembly, and grounded generation are each engineered stages — accuracy designed in, not hoped for.
- **Grounded & permissioned by construction.** A document-level authorization prefilter pushed into the search, plus answer-only-from-context and mandatory citations, means answers come only from authorized sources — and always show their work.
- **A graph that improves retrieval.** Two-phase LLM extraction with cross-document dedup and synonym merging builds a living knowledge graph that both visualizes connections and sharpens search.
- **Cost-and-latency engineering.** Five query-understanding calls collapsed into one; cheaper models for routing and the flagship reserved for reasoning; parallel retrieval; token-budget packing that preserves fusion order.
- **A meeting copilot with an adversarial panel.** Advocate, critic, arbiter, and risk agents argue a recommendation before it reaches the room, alongside live fact-checking and an LLM-scripted TTS recap.
- **Observable, measurable, scalable.** OpenTelemetry tracing across gateway and workers, per-response token-and-entity metrics, offline evaluation tooling, and independently scaling queue-driven workers on Kubernetes.

**Note on scope.** Capabilities and technique parameters reflect the delivered system. Client identity, product names, brand assets, proprietary prompts, business rules, model deployment names, credentials, and commercial metrics are excluded under NDA; outcomes here are described qualitatively.